Constraints

Technical Constraints

- Time: 5-week school project timeline (iterative implementation required)
- Complexity: Must be demonstrable/testable within academic context (likely simulated integrations)
- Infrastructure: No access to real procurement systems (requires mock data and APIs)
- Scale: Small manufacturing company context (10-50 SKUs, 5-10 suppliers realistic scope)
- Integration: Email, inventory systems, supplier APIs must be mocked or stubbed

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Functional Constraints

- Autonomy boundaries: Agent cannot operate completely unsupervised (human approval workflow required)
- Financial limits: Maximum autonomous spend amount must be defined
- Supplier limitations: Pre-approved supplier list (agent doesn't find new suppliers autonomously)
- Scope boundaries: Focus on material procurement, not services/contracts/capital equipment

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Performance Constraints

- Success metrics: Must prevent 2+ stockouts, catch 1+ invoice error, achieve 95%+ approval rate
- Response time: Stockout detection must occur with enough lead time for delivery
- Learning speed: Agent must show improvement within demonstration timeframe (5-10 decisions)

Safety Constraints

- No catastrophic decisions: Safeguards must prevent scenarios like ordering 1000x quantity, selecting unapproved suppliers, exceeding budget by orders of magnitude
- Reversibility: Bad decisions must be identifiable and correctable
- Escalation: Edge cases beyond agent capability must escalate to human
- Transparency: All decisions must be explainable to stakeholders

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Ethical Constraints

- Fairness: No bias toward/against suppliers based on non-business factors
- Privacy: Business data must be handled appropriately (in real deployment)
- Accountability: Clear audit trail for every autonomous decision
- Human oversight: Humans remain accountable for agent actions

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Educational Constraints

- Documentation required: Decision log must track research, choices, failures, pivots
- Demonstration required: Must show working system with realistic scenarios

- Learning objective: Understanding autonomous agent architecture, not just implementation
- Reflection required: Ethics section on safeguards and what could go wrong